xMind implementation status

Core runtime and execution

22 October 2026 | Completed native checkpoint: d5b4231e | Evidence: bc1a9860

Area	What we implemented	What is verified / still left
C++ backend and SQLite	The backend owns sessions, conversations, run state, settings, credentials, approvals and events. Database access goes through embedded xlang3 into SQLite. Browser, CLI and VS Code share these backend contracts.	The last complete checkpoint passed 172/172 native tests. The newer permission and resume changes are still being validated.
xlang3 IPC and workers	xMind registers its root object at startup and exposes service subobjects. Separate xMind worker processes call their methods through standard xlang3 SDK IPC, following Cantor. xlang3 handles remote references and serialization.	Cross-process execution, cancellation, object lifetime and worker retirement have tests. The intermittent xlang3 runtime crash remains unresolved.
Coding tools	Workspace reads and search; file creation, editing, patching and restoration; process execution; skills; and language-server queries. Mutations use backend permissions, exact proposals and approval records.	Controlled browser and CLI tasks completed real reads, approved edits and actual Node checks. Broader everyday coding reliability still needs acceptance.
Agents, graphs and planning	A normal agent chooses tools dynamically. Graphs provide predefined dependencies. Planning creates agent, tool and human tasks. Delegated children share the root budget, deadline and cancellation, with separate histories.	Main process roots support planning, delegation and human pause/reopen. Nested planning inside dynamic process children is unfinished. Permission binding across resume is being validated.
Model providers	Native provider adapters support saved model profiles, credential references, streaming, tool calls, usage and provider-specific conversation history.	Controlled protocol tests cover several providers. They do not establish live acceptance for every advertised model. Development checks currently avoid paid inference.

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Clients, interoperability and delivery

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Area	What we implemented	What is verified / still left
MCP and A2A	MCP connects external tools. A2A supports agent discovery and task communication. Operations pass through backend authorization, approvals and durable result recording.	Controlled interoperability tests pass. Wider testing against independent real servers and agents remains.
Browser UI	Workspace selection, scoped history, conversation streaming, file browsing, approvals, planning controls and the conversation/footer layout.	Candidate checks verified workspace switching, coding, reload and footer behavior. The latest changes have not reached the installed version.
CLI	Commands use the same backend for sessions, runs, history, approvals, events and watching progress. Managed profiles discover and attach to a backend.	Packaged coding passed through an explicit backend port. Managed-profile lifecycle tests also pass. Final packaged discovery and upgrade acceptance remains.
VS Code	A thin extension provides the workspace pane, conversation, approvals and editor integration. Execution and persistence stay in the C++ backend.	Automated adapter tests pass. The latest package still needs actual VS Code activation and rendered coding verification.
Packaging and installation	A versioned bundle contains the native executable, pinned xlang3 runtime and modules, allowed libraries and license notices.	A candidate was staged. The final update needs qualification and a protected installation that preserves settings, workspaces and history.